

Group Project Datasets - Divergence Analysis Summary

H524 Fall 2025 - Final Dataset Selection

Date: November 8, 2025 **Objective:** Maximize AI divergence (target: 8+/10 for all groups)

Final Dataset Lineup

Group	Dataset	N	Divergence Score	Key Divergence Feature
1	Anorexia Treatment	72	8/10	Pre-post design (ANOVA vs ANCOVA vs paired tests)
2	ToothGrowth	60	8/10	Factorial 2×3 design (interaction effects)
3	Bacteria	220	9/10	Repeated measures (independence trap!)
4	Pima Diabetes	200	7/10	Binary outcome, multiple predictors

Average Divergence: 8.0/10 Result: 3 of 4 groups have Very High divergence (8+/10)

Group 1: Anorexia Treatment Study

File: group_1/anorexia.csv **Design:** Pre-post intervention with 3 treatment groups **Outcome:** Weight (continuous, pounds)

Divergence Score: 8/10 (Very High)

Why High Divergence?

Multiple valid analytical approaches: 1. **ANOVA on change scores** (Postwt - Prewt) - Simple, intuitive 2. **ANCOVA** (Postwt ~ Treat + Prewt) - More rigorous, adjusts for baseline 3. **Paired t-tests** within each group - Misses between-group comparison 4. **Repeated measures ANOVA** - Advanced but appropriate

Sample Prompts & Expected AI Differences

Prompt: “Analyze this anorexia treatment data”

- **Claude (predicted):** ANOVA on change scores, accessible narrative, minimal assumptions
- **ChatGPT (predicted):** ANCOVA with baseline adjustment, rigorous assumption checking, effect sizes
- **Copilot (predicted):** Code-focused, may suggest simple approach

Actual Results from Testing

Approach	Method	p-value	Conclusion
A	ANOVA on change scores	0.0065	Significant
B	ANCOVA (baseline-adjusted)	0.0002	Significant (stronger)
C	ANOVA on final weight only	0.0004	Significant

Key Finding: Different p-values depending on method choice!

Critical Decision Points

1. What to analyze? (change vs final vs baseline-adjusted)
2. Check normality assumptions or not?
3. Include post-hoc tests or not?
4. Report effect sizes or not?

Student Assignment

Group 1 Students: 1, 4, 10, 12, 13 (n=5)

Group 2: ToothGrowth Study

File: group_2/toothgrowth.csv **Design:** 2×3 factorial experiment (Supplement × Dose) **Outcome:** Tooth length (continuous, mm)

Divergence Score: 8/10 (Very High)

Why High Divergence?

Factorial design creates analytical complexity: 1. **Two-way ANOVA with interaction** - Proper factorial analysis 2. **Two-way ANOVA main effects only** - Simpler but misses interaction 3. **One-way ANOVA** (6 groups) - Treats as simple comparison 4. **Linear regression** (dose as continuous) - Different approach 5. **Separate t-tests** - Multiple comparisons issue

Sample Prompts & Expected AI Differences

Prompt: “Compare tooth growth across supplement types and doses”

- **ChatGPT (predicted):** Two-way ANOVA with interaction, full factorial analysis
- **Claude (predicted):** One-way ANOVA or simpler approach
- **Copilot (predicted):** May miss factorial structure

Actual Results from Testing

Approach	Interaction Term	Interaction p-value	Finding
A	Yes (supp × dose)	0.0219	Significant interaction!
B	No (main effects only)	N/A	Misses complexity
C	Ignored (t-test only)	N/A	Wrong design recognition

CRITICAL: Significant interaction (p=0.0219) means supplement effect **depends on dose!** AI that ignores interaction misses important finding.

Critical Decision Points

1. Recognize factorial structure or not?
2. Include interaction term or not?
3. Treat dose as categorical or continuous?
4. How to interpret significant interaction?

Student Assignment

Group 2 Students: 7, 11, 14, 15, 16 (n=5)

Group 3: Bacteria Treatment Study HIGHEST DIVERGENCE!

File: group_3/bacteria.csv **Design:** Repeated measures (50 people, multiple time points each) **Outcome:** Bacteria presence (binary: yes/no)

Divergence Score: 9/10 (VERY HIGH - Best Found!)

Why VERY High Divergence?

REPEATED MEASURES TRAP: - Same 50 individuals measured at weeks 0, 2, 4, 6, 11 - 220 total observations but only 50 **independent** units - Standard chi-square/logistic regression **VIOLATES independence assumption** - Naive analysis gives **WRONG results!**

Sample Prompts & Expected AI Differences

Prompt: “Does treatment affect bacteria presence?”

- **Naive AI:** Simple chi-square on all 220 observations (**WRONG!**)
- **Intermediate AI:** Logistic regression ignoring ID (**WRONG!**)
- **Advanced AI:** Recognizes repeated measures, suggests GEE/mixed models (beyond intro level)
- **Clever AI:** Person-level summary, then chi-square (**CORRECT and intro-appropriate!**)

Actual Results from Testing - THIS IS AMAZING!

Approach	Method	p-value	Conclusion	Correct?
A	Chi-square (naive)	0.0358	Significant	WRONG!
B	Person-level summary	0.4232	NOT significant	CORRECT
C	Final timepoint only	0.6133	NOT significant	CORRECT

CRITICAL FINDING:

Naive analysis: $p=0.0358 \rightarrow$ “Treatment works!” (WRONG conclusion) **Correct analysis:** $p=0.4232 \rightarrow$ “No evidence treatment works” (CORRECT conclusion)

Completely opposite scientific conclusions!

Why This Is Perfect for the Project

1. **Tests AI data structure awareness:** Does AI check for repeated measures?
2. **Wrong approach is tempting:** Naive chi-square seems reasonable at first
3. **Intro-appropriate solution exists:** Person-level summary is understandable
4. **Teaches critical concept:** Independence assumption in statistics
5. **Distinguishes AI sophistication:** Rigorous AI explores data first, naive AI jumps to analysis

Critical Decision Points

1. **CRITICAL:** Will AI recognize repeated measures structure?
2. How to handle repeated measures (summary vs final vs wrong)?
3. Which summary statistic (ever positive? final only? proportion positive)?
4. Understanding why naive approach gives anti-conservative p-values

Student Assignment

Group 3 Students: 3, 5, 8, 18 (n=4)

Group 4: Pima Diabetes Study

File: group_4/pima_diabetes.csv **Design:** Cross-sectional study **Outcome:** Diabetes (binary: Yes/No)

Divergence Score: 7/10 (High)

Why High Divergence?

Multiple predictors and method choices: 1. **Chi-square tests** (simple, one predictor at a time) 2. **t-tests** (using continuous predictors like glucose) 3. **Logistic regression** (multivariate) 4. Binary vs continuous outcome analysis

Sample Prompts & Expected AI Differences

Prompt: "What factors predict diabetes?"

- **Simple AI:** Chi-square tests for each predictor separately
- **Intermediate AI:** Multiple t-tests
- **Advanced AI:** Multivariate logistic regression

Actual Results from Testing

Approach	Method	Glucose p-value	Finding
A	Chi-square (glucose high/low)	0.0000	Significant
B	t-test (glucose continuous)	0.0070	Significant
C	Logistic regression (multivariate)	<0.001	Significant

All methods agree on significance, but strength varies.

Critical Decision Points

1. Binary outcome (diabetes Yes/No) vs continuous predictors?
2. Chi-square vs t-test vs logistic regression?
3. Univariate vs multivariate analysis?
4. Which predictors to include?

Student Assignment

Group 4 Students: 2, 6, 9, 17 (n=4)

Divergence Score Distribution

9/10: 1 dataset (25%) - Bacteria
8/10: 2 datasets (50%) - Anorexia, ToothGrowth
7/10: 1 dataset (25%) - Pima Diabetes

Mean: 8.0/10

Median: 8.0/10

Result: 75% of groups have Very High divergence (8+/10)

Pedagogical Value by Dataset

Anorexia (Group 1)

Teaches: Pre-post analysis, baseline adjustment, change scores, ANCOVA

ToothGrowth (Group 2)

Teaches: Factorial designs, interaction effects, two-way ANOVA

Bacteria (Group 3)

Teaches: Independence assumption, repeated measures, data structure exploration **WHY THIS MATTERS:** Most important statistical assumption students often ignore!

Pima Diabetes (Group 4)

Teaches: Binary outcomes, multiple predictors, method selection for categorical data

What This Tests About AI Tools

Data Structure Awareness

- **Bacteria:** Does AI recognize repeated measures before analyzing?
- **ToothGrowth:** Does AI recognize factorial structure?
- **Anorexia:** Does AI recognize pre-post paired design?

Statistical Rigor

- **All datasets:** Does AI check assumptions (normality, independence)?
- **Bacteria:** Does AI understand why naive approach is wrong?
- **ToothGrowth:** Does AI test for interaction effects?

Method Selection

- **Anorexia:** ANOVA vs ANCOVA decision
- **ToothGrowth:** One-way vs two-way ANOVA
- **Bacteria:** Repeated measures handling
- **Pima:** Chi-square vs logistic regression

Interpretation Quality

- Does AI explain **WHY** a method is chosen?
 - Does AI report effect sizes or just p-values?
 - Does AI provide context-appropriate interpretation?
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Validation Complete

All 4 datasets tested with: - Multiple sample prompts (vague → specific) - Multiple valid statistical approaches run in R - Actual p-values and results documented - Decision points identified where AI will diverge - Intro-appropriateness confirmed

Files created: - group_1/divergence_test_anorexia.R - Full testing script - group_2/divergence_test_toothgrowth.R - Full testing script - group_3/divergence_test_bacteria.R - Full testing script - group_4/divergence_test_pima.R - Full testing script

Project Success Criteria Met

High divergence: 3 of 4 datasets score 8+/10 **Diverse designs:** Pre-post, factorial, repeated measures, cross-sectional **Diverse outcomes:** Continuous (2), binary (2) **Intro-appropriate:** All analyses can be done with Week 1-8 methods **Real medical data:** All from peer-reviewed studies **Pedagogically valuable:** Each teaches different critical concept **AI comparison viable:** Confirmed divergence through actual testing

Key Takeaway

The Bacteria dataset (Group 3) is the crown jewel: - Highest divergence score (9/10) - Tests most critical statistical concept (independence) - Wrong analysis gives **opposite conclusion** - Perfect for demonstrating why AI verification is essential

This dataset alone justifies the entire AI comparison project!

STATUS: Dataset selection finalized and validated **NEXT STEP:** Create instructor guides and Week 7 assignment materials